

UMAR ALI

AI Engineer | Generative AI & RAG Specialist | Machine Learning & Deep Learning
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PROFESSIONAL SUMMARY

Certified AI Engineer specializing in Generative AI, RAG Systems, Machine Learning, and Deep Learning with 2+ years of experience building production-ready intelligent applications. Proven track record designing and deploying scalable RAG-based chatbots using LangChain and LangGraph, developing CNN architectures for image classification (90%+ accuracy), and creating end-to-end ML pipelines with FastAPI and Flask. Proficient in TensorFlow, Keras, PyTorch, and modern LLM tools, delivering impactful solutions across healthcare, education, and recommendation systems.

TECHNICAL SKILLS

Programming Languages: Python, SQL, JavaScript, Java, C++, C#

Databases & Backend: MongoDB, Node.js, REST APIs

Generative AI & RAG: LangChain, LangGraph, RAG Systems, Vector Databases, Prompt Engineering, LLMs

ML/DL Frameworks: TensorFlow, Keras, PyTorch, Scikit-learn, OpenCV, XGBoost

Deployment & DevOps: FastAPI, Flask, Docker, Streamlit, Railway, GitHub CI/CD

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Feature Engineering, Statistical Analysis

ML Techniques: CNN, RNN, Transfer Learning, Clustering, Regression, Classification, NLP, Ensemble Methods

Specializations: Generative AI, RAG Systems, Conversational AI, Healthcare AI, Image Classification, Recommender Systems

PROFESSIONAL EXPERIENCE

AI/ML Engineer – Freelance & Independent Projects | 2022 – Present

- Developed 15+ production-grade AI/ML solutions specializing in GenAI, RAG systems, and deep learning.
- Built conversational AI chatbots using LangChain and LangGraph with context-aware retrieval mechanisms.
- Deployed scalable ML models using FastAPI and Flask with RESTful API endpoints for real-time inference.
- Engineered end-to-end ML pipelines including data preprocessing, model training, evaluation, and deployment.
- Implemented CNN architectures for image classification achieving 90%+ accuracy on healthcare datasets.
- Collaborated with stakeholders to define requirements and deliver AI solutions aligned with business goals.

KEY PROJECTS

Smart AI Notes System | github.com/Umarali8627/smart-ai-notes

- Built an intelligent note-taking application with AI-powered summarization and organization capabilities.
- Integrated LangChain for automatic content analysis, tag generation, and context-aware note management.
- Deployed with FastAPI backend featuring real-time processing and an intuitive user interface.

Conversational Chatbot with Memory & Context Retention | github.com/Umarali8627/chatbot-memory

- Developed intelligent chatbot with persistent conversational memory using LangChain memory modules.
- Implemented context retention across sessions enabling personalized, coherent multi-turn conversations.
- Deployed live on Streamlit (chatbot-with-memory.streamlit.app) with real-time response generation.

University Admission RAG Chatbot | github.com/Umarali8627/admission-chatbot

- Developed a Retrieval-Augmented Generation chatbot for university admission inquiries using LangChain and LangGraph.
- Implemented vector database for semantic search with optimized query response time under 2 seconds.
- Engineered custom preprocessing pipeline for university documentation, improving retrieval accuracy by 35%.

Eye Disease Classification System (EYE LAB) | github.com/Umarali8627/Eye_disease_Classifier_VGG

- Designed deep learning solution for real-time fundus image classification using CNN (VGG19, MobileNetV2).

- Achieved 92% classification accuracy through transfer learning and data augmentation techniques.
- Deployed as Flask web application with REST API for clinical use, processing 50+ predictions per minute.

Movie Recommendation System | github.com/Umarali8627/Movie-Recommendation-System

- Built collaborative filtering recommender system using TMDB 5000 Movies dataset with Scikit-learn.
- Deployed full-stack application on Railway with Streamlit frontend and CI/CD via GitHub integration.
- Implemented content-based filtering with cosine similarity for personalized recommendations.

Heart Disease Prediction with Ensemble ML | github.com/Umarali8627/Cardio-Project

- Developed predictive models for cardiovascular disease using ensemble methods (Random Forest, XGBoost).
- Achieved 88% accuracy on UCI heart disease dataset through feature engineering and hyperparameter tuning.
- Performed comprehensive EDA with Matplotlib/Seaborn visualizations for model interpretability.

Email Spam Detection with NLP | github.com/Umarali8627/spam_Detection

- Implemented text classification system using Multinomial Naive Bayes with TF-IDF vectorization.
- Applied NLP preprocessing: tokenization, stop-word removal, and feature extraction.
- Achieved 96% classification accuracy on email spam dataset with an optimized pipeline.

EDUCATION

Bachelor of Science in Computer Science

Northern University, Nowshera | **CGPA: 3.83 / 4.0**

FSc Pre-Medical

Hadaf Group of Colleges, Risalpur | **Grade: A**

CERTIFICATIONS & ACHIEVEMENTS

- AI/ML & Data Science Professional Certification – Ehunar (6-Month Intensive Program)
- Deep Learning Specialization – SuperDataScience
- 1st Place – Speed Programming Competition, Abasyn University Peshawar
- 1st Place – Startup Competition, Dawn Fest Mardan 2025